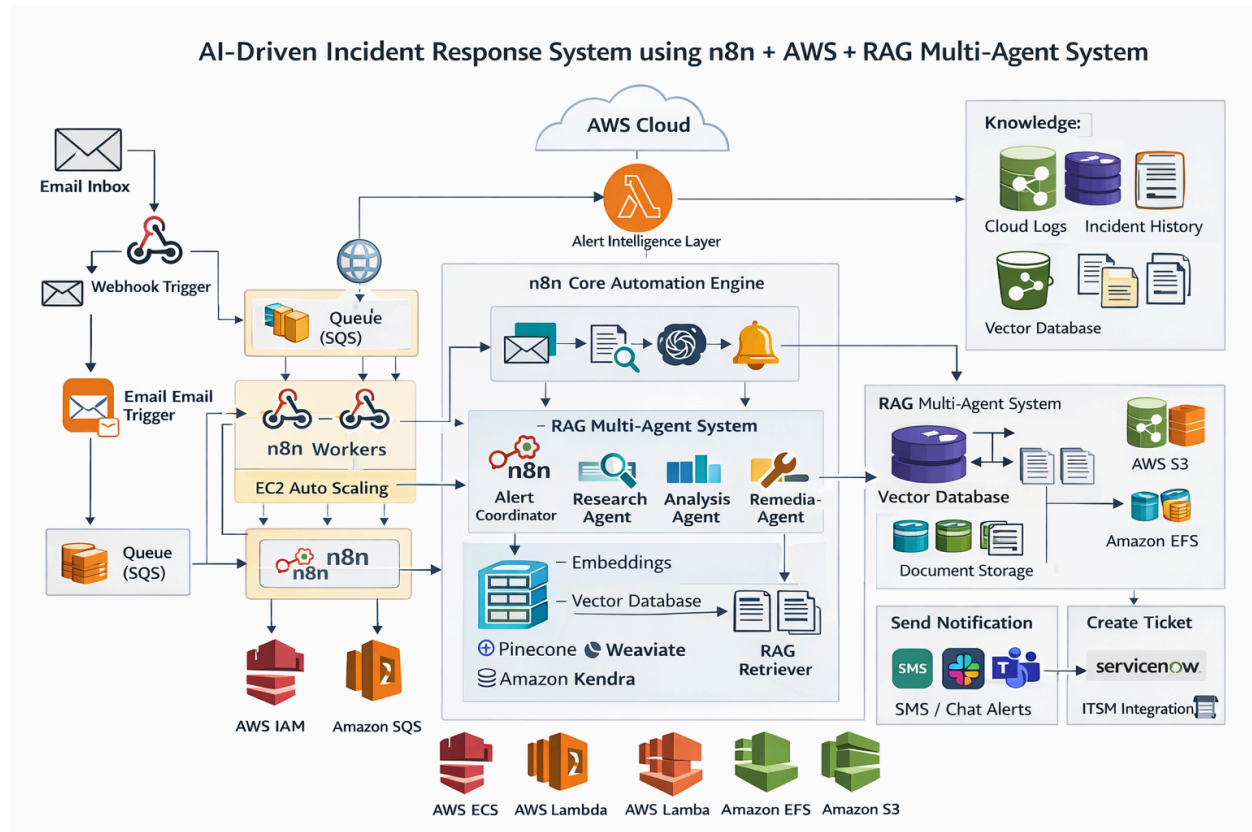




1. Email Inbox (Alert Source)

Concept

The **Email Inbox** is the **initial event source** of the system. Monitoring tools send alerts via email when infrastructure issues occur.

Examples of alert sources:

- AWS CloudWatch
- Datadog
- Prometheus AlertManager
- Jenkins
- Security tools
- Application monitoring systems

Example alert:

Subject: CRITICAL ALERT: EC2 CPU > 95%

Instance: prod-api-server
CPU: 97%
Region: us-east-1
Time: 02:12 AM

Role in Architecture

Monitoring Tools



Alert Email



Email Inbox

The inbox acts as the **trigger point for automation**.

2. Webhook Trigger

Concept

The **Webhook Trigger** is the **entry point for the automation system**.

Instead of manually checking email, alerts are forwarded to **n8n via webhook**.

Email System



Webhook Endpoint



n8n Workflow

Example endpoint:

<https://automation.company.com/webhook/alert>

Role

- Receives alert payload

- Starts automation workflow
- Converts email into structured data

n8n Node

Webhook Node

Example payload:

```
{  
  "subject": "CRITICAL: CPU Spike",  
  "body": "Instance prod-db-01 CPU 97%"  
}
```

3. Email Trigger (Alternative to Webhook)

Some systems **poll mailbox directly**.

Instead of forwarding emails, n8n checks inbox periodically.

Node

IMAP Email Trigger

Architecture:

```
Mailbox  
↓  
IMAP Trigger  
↓  
n8n
```

Benefits:

- Easy integration
 - No webhook required
-

4. Amazon SQS (Queue Layer)

Concept

Amazon SQS provides **asynchronous processing and buffering**.

Queues prevent overload when **many alerts arrive simultaneously**.

Why Queue?

Imagine:

10,000 alerts/minute

Without queue:

n8n crashes

With SQS:

alerts buffered safely

Architecture:

Webhook



SQS Queue



n8n Workers

Benefits

- load balancing
 - retry handling
 - system decoupling
 - scalability
-

5. n8n Core Automation Engine

Concept

This is the **central orchestration platform** of the entire system.

n8n performs:

- workflow automation
- AI orchestration
- integrations
- event routing

Responsibilities

Parse alerts

Trigger AI analysis

Run decision rules

Trigger notifications

Create incident tickets

Core Workflow Structure

Alert Event



Data Processing



AI Analysis



Decision Logic



Incident Handling

6. n8n Workers (Auto Scaling)

Concept

Workers process queued alerts.

Architecture:

SQS Queue



Multiple n8n Workers



AI Processing

Workers run on:

AWS EC2

AWS ECS

Kubernetes

Why Workers?

Allows **horizontal scaling**.

Example:

10 workers → 10 alerts/sec

100 workers → 100 alerts/sec

7. EC2 Auto Scaling Layer

Concept

Automatically increases or decreases **n8n worker instances** based on workload.

Example rule:

Queue depth > 500

→ launch new EC2 instance

Architecture:

High Alert Load
↓
Auto Scaling
↓
Launch More Workers

8. Alert Intelligence Layer

This is the **AI processing layer**.

It converts raw alerts into **meaningful insights**.

Tasks performed:

Alert classification
Root cause analysis
Incident grouping
Alert summarization
Remediation suggestions

9. RAG Multi-Agent System

This is the **most advanced part of the architecture**.

RAG = **Retrieval Augmented Generation**

AI combines:

LLM reasoning
+ enterprise knowledge

Multi-Agent Structure

The system contains specialized agents:

1. Alert Coordinator Agent

Responsibilities:

Receive alert
Determine analysis workflow
Assign tasks to agents

2. Research Agent

Searches knowledge sources:

logs
incident history
documentation
runbooks

3. Analysis Agent

Performs:

root cause analysis
pattern recognition
impact assessment

Example:

CPU spike + memory leak
→ caused by worker process

4. Remediation Agent

Suggests or executes solutions.

Examples:

restart service
scale EC2 instance

clear cache
roll back deployment

10. Embeddings Layer

Embeddings convert text into **numerical vectors**.

Example:

"CPU spike in database"

↓

vector representation

This enables **semantic search**.

Used for:

similar incident detection
knowledge retrieval
alert correlation

11. Vector Database

The **vector database** stores embeddings of:

alerts
logs
runbooks
incident reports
solutions

Popular technologies:

Pinecone
Weaviate
Qdrant

Chroma
Amazon Kendra

Example

New alert:

Database CPU spike

Vector search finds:

similar incident 3 weeks ago
solution: restart worker

12. Knowledge Layer

Stores enterprise knowledge.

Includes:

Cloud logs
Incident history
Operational runbooks
Architecture documentation

Sources:

CloudWatch Logs
S3 documents
Confluence
Internal documentation

13. Document Storage

Documents used for RAG.

Examples:

runbooks
architecture diagrams
incident reports
support documentation

Storage:

Amazon S3
Amazon EFS
internal storage

14. Amazon S3

S3 is used for:

log archives
alert records
training data
incident reports
documents

Benefits:

cheap
scalable
durable

15. Amazon EFS

Elastic File System used for:

shared storage

document indexing
vector training data

Used by multiple services simultaneously.

16. AWS Lambda

Lambda provides **serverless compute**.

Used for:

log parsing
metric enrichment
remediation scripts
data transformation

Example:

restart EC2 instance

17. AWS IAM

IAM manages **permissions and security**.

Example roles:

n8n → access SQS
n8n → read S3
n8n → invoke Lambda

Security principle:

least privilege access

18. Notification System

Once incident is analyzed, system sends alerts.

Channels:

Slack
Microsoft Teams
SMS
Email
PagerDuty

Example message:

CRITICAL INCIDENT

Service: API Gateway
Issue: CPU 97%
Root Cause: Worker overload

Recommended Action:
Restart worker cluster

19. ITSM Integration (ServiceNow)

For serious incidents, the system automatically creates **IT service tickets**.

Platforms supported:

ServiceNow
Jira Service Management
Zendesk
Freshservice

Example ticket:

Incident: API CPU Spike
Priority: P1
Assigned: DevOps Team

Status: Open

20. AWS ECS (Container Runtime)

ECS runs containerized services such as:

n8n
AI agents
RAG services

Benefits:

container orchestration
auto scaling
service discovery

21. Cloud Logs

Logs are critical for incident analysis.

Sources include:

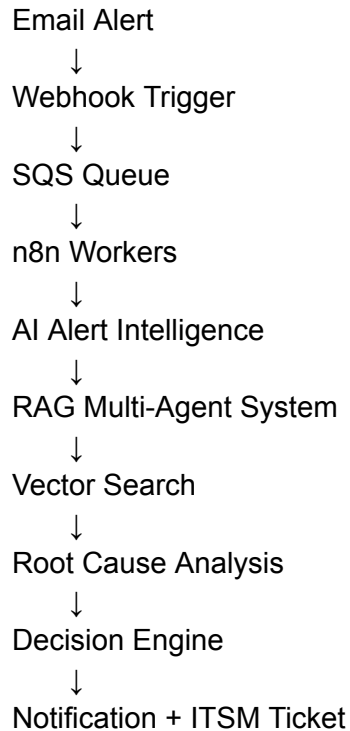
CloudWatch Logs
Application logs
Security logs
System logs

These logs feed into the **AI analysis system**.

Complete System Flow

Monitoring System





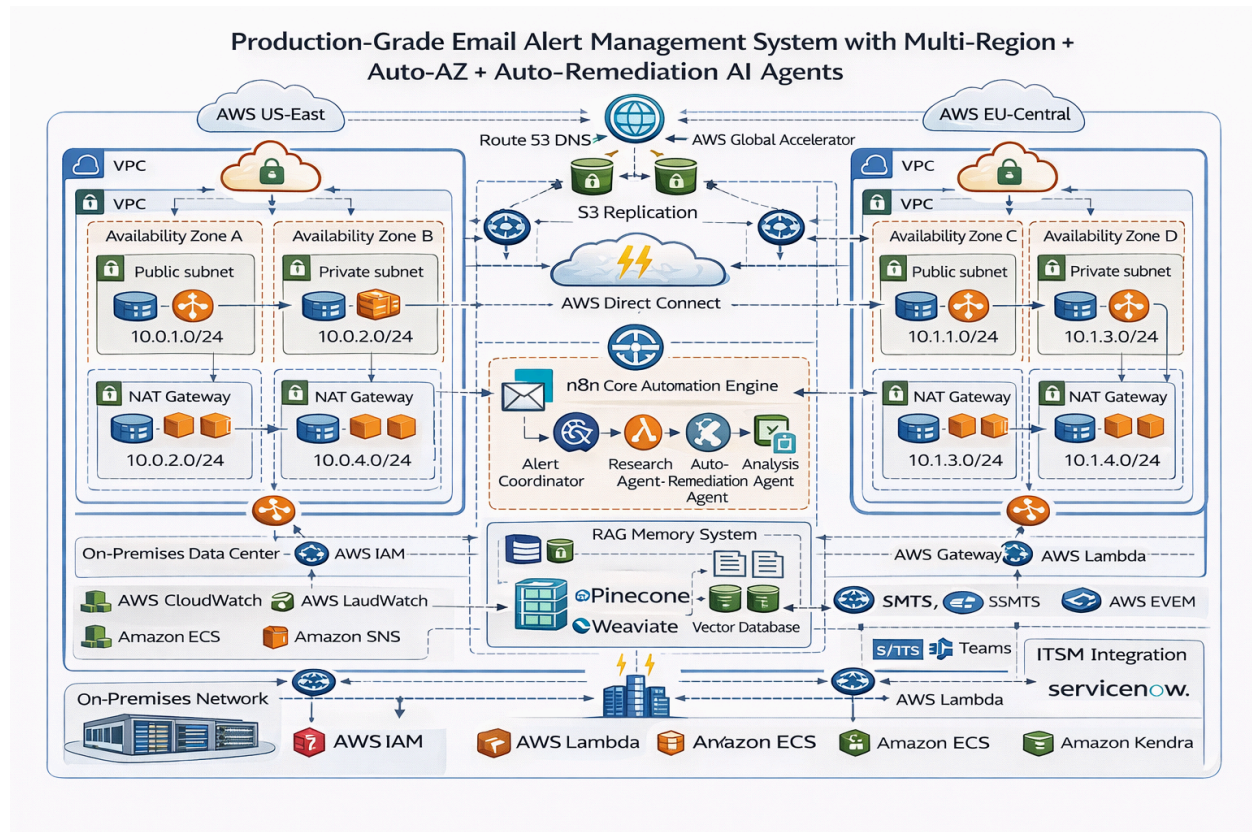
Real Enterprise Benefits

Companies handling **thousands of alerts per day** use systems like this to:

- Reduce alert fatigue
- Automate incident response
- Detect root causes faster
- Improve MTTR

Results typically include:

- 90% alert noise reduction
- 70% faster incident resolution



We will break it into **10 architectural layers** so it is easier to understand.

1. Global Traffic Layer (DNS + Global Routing)

At the very top of the architecture you see:

- **Route 53 DNS**
- **AWS Global Accelerator**

These provide **global traffic management**.

Purpose

Route alert traffic to the **closest healthy region**.

Flow

Alert Sources



Route 53 DNS



AWS Global Accelerator



Active AWS Region

What happens here

1. DNS resolves the endpoint
2. Global Accelerator selects the **lowest latency region**
3. Traffic is routed to **US-East or EU-Central**

Benefits

- Multi-region failover
 - Reduced latency
 - Disaster recovery
-

2. Multi-Region Architecture

The system runs in **two AWS regions**.

Region 1

AWS US-East

Region 2

AWS EU-Central

Why Multi-Region?

Production systems require:

- Disaster recovery
- Geographic redundancy
- Global availability

Failover

If **US-East** fails:

Route53 → EU-Central

Traffic is automatically redirected.

3. VPC Network Architecture

Each region contains its own **Virtual Private Cloud (VPC)**.

Example CIDR:

10.0.0.0/16

Purpose

The VPC provides:

- private networking
- IP addressing
- isolation
- security boundaries

Each region has its own **isolated network**.

4. Multi-Availability Zone Deployment

Inside each VPC you see **two Availability Zones**.

Example:

US-East Region

Availability Zone A

Availability Zone B

EU-Central Region

Availability Zone C

Availability Zone D

Why Multi-AZ?

AZs are **separate physical datacenters**.

Benefits:

- High availability
- Fault tolerance
- Zero downtime deployment

If **one AZ fails**, the system continues running.

5. Subnet Architecture

Each AZ contains **two types of subnets**.

Public Subnet

Example:

10.0.1.0/24

10.1.1.0/24

Public subnet hosts:

- load balancers
- NAT gateways
- internet gateways

Purpose:

receive external traffic

Private Subnet

Example:

10.0.2.0/24

10.1.3.0/24

Private subnet hosts:

- application services
- AI agents
- databases
- n8n workers

These resources are **not exposed to the internet**.

6. NAT Gateway Layer

Each AZ contains a **NAT Gateway**.

Purpose:

Allow private resources to access internet services.

Example:

n8n worker → call OpenAI API

Flow:

Private subnet



NAT Gateway



Internet

Benefits:

- secure outbound traffic
- no inbound exposure

7. Hybrid Connectivity (On-Premise Integration)

The architecture connects **enterprise datacenters** to AWS.

You can see:

On-Premises Data Center
On-Premises Network

Connected via:

AWS Direct Connect

Direct Connect

Direct Connect provides:

- private connection
- low latency
- secure networking
- predictable bandwidth

Example use cases:

send monitoring alerts
sync logs
send incident data

Architecture:

On-Prem Datacenter
↓
AWS Direct Connect
↓
AWS VPC

8. Core Automation Platform (n8n)

At the center of the architecture is the **n8n Core Automation Engine**.

This is the **orchestration platform**.

Role

n8n performs:

- email alert ingestion
 - automation workflows
 - AI orchestration
 - remediation actions
 - incident routing
-

Workflow Pipeline

Alert Email



Alert Coordinator



Research Agent



Analysis Agent



Auto-Remediation Agent

n8n coordinates **multiple AI agents**.

9. AI Agent Architecture

The system includes **AI-driven incident response agents**.

Alert Coordinator Agent

First agent.

Responsibilities:

receive alert
classify incident
route to other agents

Example:

CPU Spike

Coordinator triggers analysis.

Research Agent

Research agent gathers information.

Sources:

logs
runbooks
incident history
monitoring data

Example:

search similar incidents

Analysis Agent

This agent performs:

root cause analysis
impact assessment
severity classification

Example output:

Root cause: memory leak

Service affected: API gateway

Severity: critical

Auto-Remediation Agent

This is the **self-healing system**.

It can perform automated fixes.

Examples:

restart EC2 instance

scale ECS service

clear cache

rollback deployment

Actions executed through:

AWS Lambda

AWS APIs

ECS automation

10. RAG Memory System (AI Knowledge Layer)

This system uses **Retrieval Augmented Generation (RAG)**.

Purpose:

Give AI agents **organizational memory**.

Vector Databases

The diagram shows:

Pinecone
Weaviate
Vector Database

These store embeddings of:

past alerts
incident reports
runbooks
logs
documentation

How RAG Works

Alert arrives:

Database CPU spike

AI converts alert to embedding.

Search vector DB.

Finds similar incident:

Incident 3 weeks ago
Solution: restart background worker

AI recommends solution.

11. Observability & Monitoring

The system uses several AWS monitoring services.

AWS CloudWatch

Monitors:

CPU usage
workflow execution
alerts processed
system health

Amazon SNS

SNS distributes alert notifications.

Example:

CloudWatch Alarm



SNS Topic



n8n

12. Serverless & Compute Layer

Automation tasks use AWS compute services.

AWS Lambda

Used for:

alert preprocessing
remediation scripts
API automation
log processing

Example:

restart EC2 instance

Amazon ECS

Runs containerized workloads.

Used for:

AI agents
n8n workers
analysis services

13. Storage & Data Layer

Amazon S3

Stores:

alert archives
incident logs
training data
attachments

S3 also supports **cross-region replication**.

S3 Replication

Replicates data across regions.

Example:

US-East S3
↓
EU-Central S3

Benefits:

- disaster recovery
- global durability

14. Security Layer

AWS IAM

IAM controls:

access permissions
API authorization
service roles

Example roles:

n8n → invoke Lambda
Lambda → read S3
SES → publish SNS

Security follows:

least privilege principle

15. ITSM Integration (ServiceNow)

The system integrates with **ServiceNow**.

Purpose:

Automatically create incident tickets.

Flow:

Critical Alert



AI Analysis



Create ServiceNow Incident

Example ticket:

INC34509

Service: Payment API

Priority: P1

16. Notification Channels

Alerts can be sent to:

Microsoft Teams

Slack

SMS

Email

PagerDuty

These are triggered through:

SNS

n8n workflow

17. Complete System Flow

End-to-end architecture:

Monitoring Systems



Alert Email



Route53



Global Accelerator



AWS Region



VPC



n8n Automation
↓
AI Agents
↓
Vector Database Search
↓
Root Cause Analysis
↓
Auto Remediation
↓
ServiceNow Ticket

18. Enterprise Benefits

This architecture provides:

High Availability

Multi-Region
Multi-AZ

Scalability

ECS containers
Lambda
n8n workers

AI-Driven Incident Response

automated diagnosis
auto remediation
knowledge reuse

Disaster Recovery

cross-region replication
failover routing

19. Real-World Scale

Systems like this typically handle:

10,000 – 100,000 alerts/day

And reduce alert noise by:

80–90%
